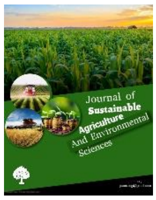
JSAES Vol(1) Issue(1) 2023

JOURNAL OF SUSTAINABLE AGRICULTURE AND ENVIRONMENTAL SCIENCES (JSAES)

Journal homepage: http://jsaes.org/index.php/jsaes/index



A COMPREHENSIVE REVIEW OF POWDERY MILDEW DISEASE IN CHILLI AND ITS MANAGEMENT STRATEGIES

MUHAMMAD ALI HASSAAN[1], HASEEB SYED[2], HANZALA OBAID[2], ZULKIFL[2], ASFA QUDDUS[2]

1 Department of Plant Pathology, Pir Mehr Ali Shah Arid Agriculture University, Rawalpindi, Pakistan
2 Department of Horticulture, Pir Mehr Ali Shah Arid Agriculture University, Rawalpindi, Pakistan

Article info

Article history:
Received: 30 Aug 2023
Accepted: 20 Sep 2023

Key words: *Capsicum annum* L., Disease Management, Epidemiology, Pest and Diseases, Powdery mildew

Hassaan *et al.*, (2023). A comprehensive review of powdery mildew disease in chilli and its management strategies: *J. S. Agri. Enviro. Sci*, *1*(1), 1-8. https://doi.org/10.5281/zenodo.8360911

https://orcid.org/0009-0006-5636-4593

***Corresponding author**
Muhammad Ali Hassaan Email : m.alihassaan786@gmail.com

Conflict of Interest: The authors declare no conflict of interest.

ABSTRACT

Capsicum annuum L., commonly known as Chilli, is a highly prized horticultural crop cultivated in various regions across the globe. In Pakistan, Chilli holds a prominent position as a major cash crop, with the Sindh province being renowned for producing high-quality Chilli. However, Chilli plants are susceptible to various pathogens, including bacteria, viruses, and fungi, leading to significant yield and quality losses. One of the most detrimental diseases affecting Chilli plants, both in open fields and under greenhouse conditions, is powdery mildew. This devastating condition is primarily caused by Leveillula taurica, also known as Oidiopsis taurica. It manifests through distinct symptoms and has a profound impact on Chilli crop production. The primary objective of this review paper is to provide an in-depth analysis of Chilli cultivation, focusing specifically on the powdery mildew disease caused by Leveillula taurica. The paper delves into the symptoms exhibited by infected plants, the epidemiology of this disease, and various management strategies employed worldwide to mitigate its impact. By understanding these critical aspects, farmers and researchers can adopt effective measures to safeguard Chilli crops and enhance their yields.

.

1. INTRODUCTION

Chili is originated in Mexico later the Columbian Exchange, many varieties of chili pepper are distributed crosswise the world, and used for both nutrient and conventional medicine (Kraft *et. al;* 2014). The chili pepper (*Capsicum annum* L.) is a native Mexican variety of chili pepper that is valued because it is the main ingredient of Oaxacan black mole, a typical Mexican dish. It is mainly grown in the Cañada region of the state of Oaxaca, Mexico, in the Tehuacán-Cuicatlán biosphere reserve. It is important to note that it is grown in conventional agricultural systems, which have a range of agronomic limitations related to production as well as the incidence and severity of pests and diseases are significant obstacles that hinder potential profits. Additionally, the genetic base of the crop is very narrow. Under such environmental and production conditions, the average yield of chili peppers can reach 1 t ha−1 dehydrated fruit, which can be used in the food, chemical and pharmaceutical industries. (García-Gaytán, *et. al;* 2017) In intense cases whitish color mass acquire both the surfaces which resulted in immature defoliation (Jharia *et. al;* 1978). Massive infection leads to shedding of leaves resultant in higher yield losses owed to decrease in size and quantity

of fruits, these necrotic lesions step by step turned in to brownish-black with the visual aspect of fungal fructify fiction stunting of plants precede by fruit driblet (Blazquez, 1976). In order to control the powdery mildew disease of chill crop many management strategies are applied. Application of chemical sprays are also effective against the disease spread.

1.1.Objective of study

This review article's main objective is to give a comprehensive analysis of the powdery mildew disease that affects chili plants. Understanding its prevalence, how it spreads, and comparing different control strategies will be our main concerns. These strategies include customs, medications, and the application of biopesticides. Our goal is to advance knowledge of powdery mildew in chili plants and encourage additional study in this area, particularly in relation to disease control strategies, through a thorough analysis of the available literature.

2. ABOUT PATHOGEN

An obligate pathogenic fungus called *Leveillula taurica* produces powdery mildew on a variety of crops, including key ones like cotton, pepper, tomatoes, eggplant, and onions. The disease's first phases are challenging to identify and can quickly spread undetected, for instance in greenhouses and fields where peppers and tomatoes are grown. A severe fungal danger to agricultural production is powdery mildew. Leveillula taurica, an endophytic white mold, is a significant disease of pepper, tomato, eggplant, onion, cotton, and other crops. It has also been seen on a variety of wild plant species. A complex problem is presented by this pathogen. The illness can spread quickly in fields and greenhouse crops due to several factors, including the fact that the early stages of infection are challenging to diagnose. Second, it is difficult to distinguish between different species within the genus Leveillula, and the binomial nomenclature "*L. taurica*" unmistakably designates a complex of species that comprises a number of different biological species. The precise host ranges of the several *L. taurica* lineages identified by phylogenetic research are yet unknown (Zheng *et. al*, 2013).

2.1.Distribution of pathogen

The Pathogen is present in almost every chilli growing region. *Leveillula taurica* was first reported from Morocco in 1937. And later in America the disease first noticed in Florida, USA in 1971. Since the early 1990's the pathogen has been continual problem in California on chili, tomato, cotton, and bell peppers, onion and other weeds species. Later in 1990's it had outspread to Arizona, Oklahoma, Utah, Idaho, Mexico, New York, and Ontario. The pathogen was firstborn reported on greenhouse pepper in 2003 in British Columbia and since dispersed throughout chilli greenhouses. The pathogen can also infect other species, including peppers, cucumbers, tomatoes and eggplants (Evans K *et. al;* 2008) Over thousands of species of plants in different families are susceptible to L. taurica.

2.2. *Symptoms of disease*

A white, powdery growth on the surface of the damaged plant portions is a common symptom of all powdery mildew fungus (Fig.1). The leaf, stem, or blossom may become completely covered in white powder when a few small spots grow larger and join together.

Fig.1 Powdery mildew on Capsicum leaves

Young succulent shoots are most susceptible to disease. The disease is common in crowded growing areas, in shady areas and in areas with limited air circulation. (Goldberg, 2003)

3. DISEASE CYCLE

The fungi responsible for causing powdery mildew in chili plants are classified as bio trophic fungi, meaning they depend on living plant cells for their survival and struggle to persist in the absence of a live host. These fungi exist in the form of perihelia or ascosporic, which also contain ascosporic. (Vielba-Fernández *et. al;* 2020)

The life cycle of powdery mildew (Fig.2) begins when conidia, tiny fungal spores, land on the leaves of chili plants. These spores germinate much like seeds, initiating growth within the leaf. Powdery mildew infects the plant and utilizes it as a source of nutrients. (Britannica, 2017) Initially, the pathogen grows invisibly within the leaf during a dormant period lasting from eighteen to twenty-one days. It then emerges through the leaf's lower surface via tiny breathing pores known as stomata, where it starts producing spores. These spores are borne individually on numerous fine stalks and strands called conidiophores. Fungal spores proliferate on the undersurface of the leaf, and germ tubes extend and branch out on the leaf's upper surface. Small structures called haustoria are formed, which allow the fungus to penetrate plant cells and absorb nutrients from the epidermal layer of these cells. (Doohan F., 2011) The majority of the fungi remain on the outside surface of the plant. On the plant's surface, new conidiophores develop. Conidiophores are structures that contain new spores, known as conidia. These conidiophores contribute to the characteristic fluffy fungal growth seen in powdery mildew infections. In the case of chili and pepper plants, infection by Leveillula taurica occurs through the stomata of leaves, and the fungus begins growing inside the leaf. Even in this situation, conidiophores extend from the natural openings, the stomata. (Zheng *et. al;* 2013)

Early summer is usually when infections happen, either because of spores transported from warmer southern regions or because of spores released from fruiting bodies in plant detritus. (Boddy L., 2016) These spores can travel great distances thanks to the wind. Powdery mildews do not grow in damp or rainy circumstances, in contrast to many fungi. For transmission to be facilitated, they just need high humidity or condensation for a short period of time. The germ tube enters the plant tissue either on the surface (as shown with Erysiphe lycopersici) or within the plant tissue (as seen with L. taurica) when the spore germinates once it settles on a suitable plant surface. The first spores are created and spread around a week after infection. Powdery mildew spreads quickly from plant to plant once spores are present in a field, helped by the wind and maybe by the clothing of workers (Kumar *et. al; 2003*).

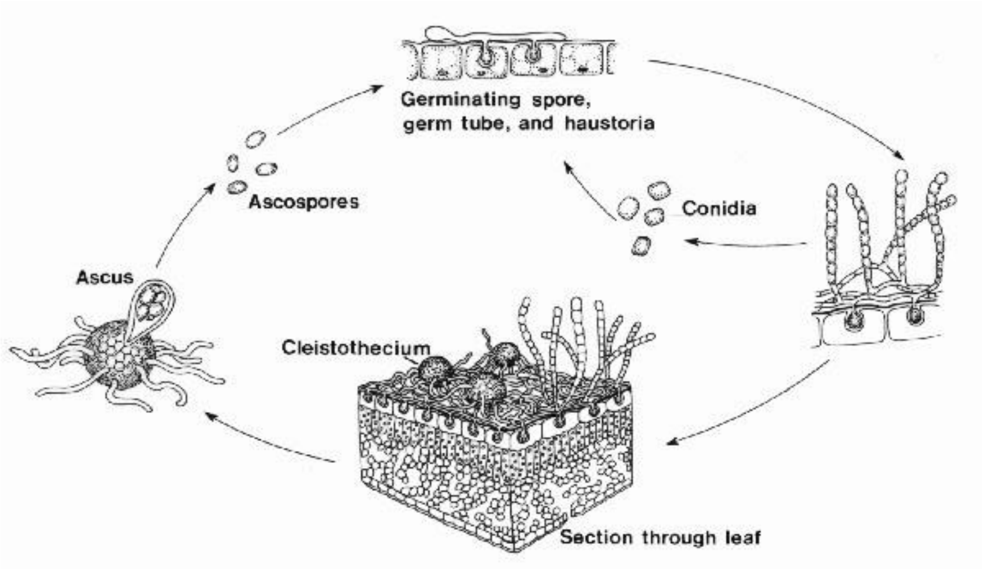


Fig. 2 Powdery Mildew Fungi: Classification and Ecology

3.1.Identification of disease in field

In field the disease is noticed at older stage when it its start producing mycleium and due to the growth of mycelium the surface of leaves of infected plants is covered by white, velvety, powdery masses which are easily identified as the plant is infected by powdery mildew (Nyvall, R. F; 2013).

4. MANAGEMENT

Powdery mildew is ruinous diseases of chilli crop, that amends the plant fruits, quality and lessen the total output. (Reddy, P. P.; 2010) As per a new report, the estimated crop loss due to this fungal disease was around 40%. Different management strategies are used to control the spread and severity of disease and these strategies are both preventive and curative.

It is difficult to control to control the disease when the whole field is infective by pathogen then the only option that's remain use of fungicide to control the disease. (McGrath; 2001)

4.1.Cultural practices

Cultural practices are also helpful in order to get rid of these pathogens some of the best cultural practices that are effective against powdery mildew includes; (Brussaard *et. al;* 2007)

i. Prune out the infected plants parts if it is possible

ii. Burn out the plant debris this reduce overwintering of fungi inoculum.
iii. Avoid excessive application of fertilizer like Nitrogen.
iv. To reduce humidity in the plant canopy, make sure to improve airflow around your plants and prune plants in crowded areas.
v. The use of pathogen-specific resistant cultivars also aids in reducing the disease. (Liang *et. al;* 2011)
vi. Exceed over watering the crop.
vii. Maintain appropriate fertilizer level.
viii. Provide adequate water.
ix. Always raise nursery in disease free environment and in healthy soil.
x. Crop rotation with non-host crop is also effective. For example, numerous studies have described that nonstop cropping reduced many organic matter and microbe's activity in the soil, foremost to the occurrence of soil borne diseases (Hanet *et. al;* 2016).

4.2. Chemical control

Chemical substances are commonly employed to control powdery mildew in chili plants. Various fungicides have been tested for their effectiveness in managing this disease.

i. Powdery mildew on chili plants has been treated with pyraclostrobin (marketed as Cabrio), azoxystrobin, and trifloxystrobin (sold as Flint). However, it has only recently been documented that these fungicides are effective against the most severe forms of the disease (Alexander and Waldenmaier, 2002). The important class of compounds known as strobilurin fungicides is extensively employed in agricultural production systems to treat a variety of fungal diseases.
ii. Sharmila et al.'s study from 2004 showed that Penconazole and Propiconazole are the two most effective treatments for powdery mildew on chili plants. Anvil 5SC (Hexaconazole), at a concentration of 0.15 percent, was also discovered by Ekbote (2006) to be extremely effective and provide the lowest illness index.
iii. According to research conducted in 2007 by Sudheendra Astaputre et al., all fungicides containing triazole compounds (such as penconazole, triademefon, hexaconazole, propiconazole, and difenconazole) were successful in decreasing powdery mildew on chili plants.
iv. Penconazole, herbal NSKE (5%), Hexaconazole, and Trichoderma harzianum (0.5%) were shown to be the most powerful and economical treatments for powdery mildew, according to (Surwase et al., 2009).
v. Alharbi et al.'s (2014) research and other studies showed that Topaz (200 EW) application considerably decreased disease severity (3.5%) and enhanced fruit production (244.5 g per plant).
vi. Azit et al.'s (2014) study found that the combi-fungicide UPF-509, which contains Azoxystrobin 8.3% and Mancozeb 66.7%, was effective at reducing powdery mildew in chili plants and produced a maximum fruit yield of 22.28 quintals per hectare when used in conjunction with other recommended fungicides.

4.3. Use of plant extract bio-pesticide

Numerous studies have explored the effectiveness of plant-extracted bio pesticides in managing diseases across various crops. The use of chemical sprays in agriculture can have adverse effects on the environment and often leads to the development of resistance in pathogens or pests (McGrath, 2001). Several plant species have been identified that naturally contain compounds in their leaves, bulbs, or shoots that are toxic to various fungi (Zhang, S. *et. al;* 2016).

i. In a field study conducted by Sudha Apswami and Prasanth Lakshmanan in 2007, the use of Carica papaya extract resulted in a significant reduction in disease incidence, with recorded percentages of 51.9%, 44.9%, and 47.3%, respectively.

ii. In a controlled environment study, extracts from A indica demonstrated high efficacy in reducing disease concentration, with a percentage of 57.2%. Following closely were the extracts of *A. cepa*, *C. papaya*, and *A. sativum*, which exhibited similar effectiveness with percentages ranging from 47.0% to 48.8% (Sudha Apswami & Prasanth Lakshmanan, 2007).

iii. Notably, extracts from *Allium spp.* bulbs, accounting for 5%, were also found to be efficient in reducing chili powdery mildew in the study conducted by Sudha Apswami and Prasanth Lakshmanan in 2007. This reduction in disease intensity may be attributed to the presence of antimicrobial substances in Allium spp. bulbs. Similar results were also reported for Erysiphe polygoni by Ramesh in 1997.

CONCLUSION

Chilli (*Capsicum annuum* L.) is a highly valued vegetable crop cultivated worldwide. Unfortunately, each year, significant yield losses occur due to the prevalence of powdery mildew disease. This disease is primarily caused by the obligate fungal pathogen Leveillula taurica, also known as Oidiopsis taurica, and it represents a major threat to chili crops. Powdery mildew significantly reduces both fruit quantity and quality, leading to substantial losses in crop yield. While managing this disease can be challenging, there are effective strategies available. Recommended fungicides, such as Penconazole, as well as plant extract bio pesticides like neem oil extract, have shown promise in controlling the severity and spread of powdery mildew in chili crops. Additionally, implementing certain cultural practices can contribute to disease management. Crop rotation, maintaining appropriate planting distances, ensuring an adequate water supply, and the removal and proper disposal of plant debris can help control the disease by eliminating its sources of inoculation. This review provides an overview of the research conducted on powdery mildew disease in chili, offering valuable insights for future studies aimed at addressing this significant agricultural challenge.

REFRENCES

1. Ajithkumar, K., Savitha, A. S., Biradar, S. A., Rajanna, B., & Ramesh, G. (2014). Management of powdery mildew and anthracnose diseases of chilli (Capsicum annuum L.). Pest Management in Horticultural Ecosystems, 20(1), 80-83.
2. Alexander, S. A., & Waldenmaier, C. M. (2002). Management of anthracnose in bell pepper. Fungicide and Nematicide Tests, 58, 49.
3. Alharbi, A. A., & Alawlaqi, M. M. (2014). Impact of some Bacillus spp., inducer resistant chemicals and cow's skim milk on management of pepper powdery mildew disease in Saudi Arabia. Life Sci. J, 11(4s), 22-28.

4. Ashtaputre, S., Kulkarni, S., Rao, M. S. L., & Shivaprasad, M. (2007). Management of powdery mildew of chilli using triazoles. J Pl Dis Sci, 2(2), 132-134.
5. BigHaat 2022. Powdery Mildew in Chilli: A Disease in Winter to Cause Heavy Crop Losses. BigHaat. https://www.bighaat.com/blogs/kb/powdery-mildew-in-chilli-a-disease-in-winter-to-cause-heavy-crop-losse
6. Blazquez, C. H. (1976). A powdery mildew of chilli caused by Oidiopsis sp. Phytopathology, 66(10), 1155-1157.
7. Boddy, L. (2016). Fungi, ecosystems, and global change. In The fungi (pp. 361-400). Academic Press.
8. Britannica, T. Editors of Encyclopaedia (2017, November 10). powdery mildew. Encyclopedia Britannica. https://www.britannica.com/science/powdery-mildew
9. Brussaard, L., Ruiter, P. C. D., & Brown, G. G. (2007). Soil biodiversity for agricultural sustainability. Agriculture, Ecosystems & Environment, 121, 233–244.
10. Chiou, K. L., Hastorf, C. A., Bonavia, D., & Dillehay, T. D. (2014). Documenting cultural selection pressure changes on chile pepper (Capsicum baccatum L.) seed size through time in Coastal Peru (7,600 BP–Present). Economic Botany, 68, 190-202.
11. Doohan, F. (2011). Fungal pathogens of plants. Fungi: biology and applications, 313-344.
12. Drost, D., Evans, K., & Frank, E. (2008). Onion Powdery Mildew.
13. Ekbote, S. D. (2006). Bio-efficacy of Anvil 5SC against chilli powdery mildew. Indian Society of Mycology and Plant Pathology, 36, 57.
14. García-Gaytán, V., Gómez-Merino, F. C., Trejo-Téllez, L. I., Baca-Castillo, G. A., & García-Morales, S. (2017). The chilhuacle chili (Capsicum annuum L.) in Mexico: Description of the variety, its cultivation, and uses. International Journal of agronomy, 2017.
15. Goldberg, N. P. (2003). Powdery mildew. Compendium of pepper diseases. APS Press: St. Paul, 19-20.
16. Han, L. L., Wang, J. T., Yang, S. H., Chen, W. F., Zhang, L. M., & He, J. Z. (2017). Temporal dynamics of fungal communities in soybean rhizosphere. Journal of Soils and Sediments, 17, 491-498.
17. Jharia, H. K., Khare, M. N., & Chand, A. (1978). Chemical control of powdery mildew of chilli. Current Research, 7, 46-48.
18. Kraft et al., (2014). Multiple lines of evidence for the origin of domesticated chili pepper, Capsicum annuum, in Mexico. Proceedings of the National Academy of Sciences, 111(17), 6165-6170.
19. Kumar, S., & De, A. K. (2003). Capsicum: historical and botanical perspectives. Capsicum: the genus Capsicum.
20. Liang, M. Q., Zhang, C. F., Peng, C. L., Lai, Z. L., Chen, D. F., & Chen, Z. H. (2011). Plant growth, community structure, and nutrient removal in monoculture and mixed constructed wetlands. Ecological Engineering, 37(2), 309-316.
21. McGrath, M. T. (2001). Fungicide resistance in cucurbit powdery mildew: experiences and challenges. Plant disease, 85(3), 236-245.
22. Nyvall, R. F. (Ed.). (2013). Field crop diseases handbook. Springer Science & Business Media.
23. Powdery Mildew of Greenhouse Peppers.(2021). https://www2.gov.bc.ca/assets/gov/farming-natural-resources-and-industry/agriculture-and-seafood/animal-and-crops/plant-health/phu-powderymildew-greenhousepeppersss.pdf
24. Powdery Mildew. (n.d.). Retrieved September 9, 2023, from https://plantclinic.nmsu.edu/documents/powdery-mildew-_od-4__final.pdf

25. Ramesh, R. (1997). Studies on peas (Pisum sativum L.) powdery mildew caused by Erysiphe polygoni.
26. Reddy, P. P. (2010). Fungal diseases and their management in horticultural crops. Scientific Publishers.
27. SARATCHANDRA, B., BISWAS, S., & DAS, N. (2002). Evaluating different plant extracts against three major diseases of mulberry.
28. Sharmila, A. S., Kachapur, M. R., & Patil, M. S. (2004). Field evaluation of fungicides against powdery mildew (Leveillula taurica (Lev.) Arn) of chilli (Capsicum annuum L.). J. Mycol. Plant Pathol, 34, 98-99.
29. Singh, U. P., Singh, H. B., & Chauhan, V. B. (1984). Effect of some fungicides, plant extracts and an oil on inoculum density of different nodal leaves of pea infected by Erysiphe polygoni/Zur Wirkung einiger Fungizide, Pflanzenextrakte und eines Pflanzenöls auf die Sporulationsintensität von Erysiphe polygoni an unterschiedlich alten Erbsenblättern. Zeitschrift für Pflanzenkrankheiten und Pflanzenschutz/Journal of Plant Diseases and Protection, 20-26.
30. Singh, U. P., Srivastava, B. P., Singh, K. P., & Mishra, G. D. (1991). Control of powdery mildew of pea by ginger extract. Indian Phytopathology.
31. Sudha, A., & Lakshmanan, P. (2007). Efficacy of botanicals against chilli powdery mildew caused by Leveillula taurica (Lev.) Arn. Madras Agricultural Journal, 94(jan-jun), 1.
32. Surwase, A. G., Badgire, D. R., & Suryawanshi, A. P. (2009). Management of pea powdery mildew by fungicides, botanicals and bio-agents. Annals of Plant Protection Sciences, 17(2), 384-388.
33. University, U.S. (n.d.). Powdery Mildew. Extension.usu.edu. https://extension.usu.edu/vegetableguide/tomato-pepper-eggplant/powdery-mildew
34. Vielba-Fernández, A., Polonio, Á., Ruiz-Jiménez, L., de Vicente, A., Pérez-García, A., & Fernández-Ortuño, D. (2020). Fungicide resistance in powdery mildew fungi. Microorganisms, 8(9), 1431.
35. Zhang, S., Mersha, Z., Vallad, G. E., & Huang, C. H. (2016). Management of powdery mildew in squash by plant and alga extract biopesticides. The plant pathology journal, 32(6), 528.
36. Zheng, Z., Nonomura, T., Bóka, K., Matsuda, Y., Visser, R. G., Toyoda, H., ... & Bai, Y. (2013). Detection and quantification of Leveillula taurica growth in pepper leaves. Phytopathology, 103(6), 623-632.